

BST 223 Data Analysis Project: Initial Model & Results

Lucas Webster

Overview

In this ‘Initial Model and Results’ portion of the project, I plan to first run a proportional-odds cumulative-logit model on every variable, and then remove those with obvious p-value, clinical, and covariance issues that were discussed in the EDA stage. Then we will rerun our model and test the proportionality assumption, to see if we can proceed with model selection and testing under this assumption. From there, I will narrow down several candidate nested models of which I will perform LR tests and use BIC/AIC to compare them. Using BIC, AIC, and these LR tests, I will narrow down to 2 models, of which I will provide coefficient interpretation for what I believe to be the ‘leader’ among them, and discuss problems I ran into along the way.

For reference, our model takes the form...

$$\log\left(\frac{P(Y_i \leq k)}{P(Y_i > k)}\right) = \alpha_k - X_i^T \beta$$

Initial Proportional Model

The below is our initial model with all covariates. I will be on the watch for kcal_stan, it had a VIF of 37 from our EDA, likely due to its high covariance with the macronutrient measures. The same goes for carb_stan, as it had a VIF above 15.

	coef	std err	z	P > z	[0.025	0.975]
cig_smoker_ever	0.0281	0.082	0.345	0.730	-0.132	0.188
sleep_weekday	-0.0521	0.026	-2.039	0.041	-0.102	-0.002
no_ins_12_mon	-0.1011	0.188	-0.537	0.591	-0.470	0.268
bmi	0.1998	0.008	26.417	0.000	0.185	0.215
sex	0.2580	0.082	3.157	0.002	0.098	0.418
age	0.0166	0.003	6.157	0.000	0.011	0.022
born_us	-0.0005	0.131	-0.004	0.997	-0.256	0.255
fam_inc_to_pov	-0.0099	0.030	-0.331	0.741	-0.068	0.049
hep_b	0.4386	0.292	1.500	0.134	-0.135	1.012
mins_seden_daily_stan	-0.0898	0.040	-2.256	0.024	-0.168	-0.012
kcal_stan	0.4812	0.228	2.108	0.035	0.034	0.929
carb_stan	0.1153	0.164	0.703	0.482	-0.206	0.437
sugar_stan	-0.1564	0.085	-1.837	0.066	-0.323	0.010
fat_stan	-0.1318	0.128	-1.029	0.304	-0.383	0.119
chol_stan	-0.0921	0.049	-1.891	0.059	-0.188	0.003
water_stan	-0.0460	0.039	-1.188	0.235	-0.122	0.030
salt_stan	0.0796	0.066	1.204	0.229	-0.050	0.209
caff_stan	-0.1123	0.039	-2.844	0.004	-0.190	-0.035
hs_dropout	0.0535	0.294	0.182	0.856	-0.524	0.631
hs_grad	0.0091	0.269	0.034	0.973	-0.518	0.536
aa	-0.1228	0.266	-0.462	0.644	-0.644	0.398
bach	-0.1304	0.273	-0.478	0.633	-0.665	0.404
prediabetes	0.1735	0.101	1.710	0.087	-0.025	0.372
diabetes	0.0685	0.122	0.559	0.576	-0.171	0.308
private_ins	-0.1373	0.085	-1.622	0.105	-0.303	0.029
rare_drink	-0.0375	0.158	-0.238	0.812	-0.347	0.272
some_drink	0.0970	0.164	0.591	0.555	-0.225	0.419
often_drink	0.1127	0.180	0.624	0.532	-0.241	0.466
drk_but_no12	0.0364	0.170	0.214	0.831	-0.298	0.370
fiber	-0.0254	0.007	-3.735	0.000	-0.039	-0.012
vit_e	-0.0298	0.009	-3.194	0.001	-0.048	-0.011
mexican	-0.2403	0.194	-1.240	0.215	-0.620	0.140
hispanic	-0.1058	0.161	-0.656	0.512	-0.422	0.211
black	-0.6134	0.135	-4.553	0.000	-0.877	-0.349
asian	0.3052	0.207	1.475	0.140	-0.100	0.711
other	-0.2435	0.159	-1.530	0.126	-0.555	0.068
0/1	4.8906	0.467	10.464	0.000	3.975	5.807
1/2	-0.3245	0.048	-6.770	0.000	-0.418	-0.231
2/3	-0.0764	0.043	-1.792	0.073	-0.160	0.007

We will make a few decisions for variable selection here. First, we will drop carb_stan, it had

an incredibly high VIF (17!) value in the EDA stage and has an insignificant p-value here. Carbs also showed little correlation in its visual diagnostic. We will also remove education level covariates, they also had incredibly high VIF in the EDA stage and p-values now, and while they showed some correlation in their visual diagnostic, common sense tells me they are likely a mask and collinear with fam_inc_to_pov, ie education level is a mask for economic status. Hence we remove them, which will also greatly increase the power of our model.

	coef	std err	z	P> z	[0.025	0.975]
cig_smoker_ever	-0.0028	0.080	-0.035	0.972	-0.160	0.154
sleep_weekday	0.0101	0.025	0.398	0.691	-0.040	0.060
no_ins_12_mon	-0.3833	0.191	-2.009	0.045	-0.757	-0.009
bmi	0.2042	0.008	26.715	0.000	0.189	0.219
sex	0.3523	0.082	4.319	0.000	0.192	0.512
age	0.0160	0.003	5.949	0.000	0.011	0.021
born_us	-0.0473	0.129	-0.368	0.713	-0.300	0.205
fam_inc_to_pov	-0.0170	0.028	-0.604	0.546	-0.072	0.038
hep_b	-0.7660	0.308	-2.486	0.013	-1.370	-0.162
mins_seden_daily_stan	0.0314	0.039	0.797	0.426	-0.046	0.109
kcal_stan	-0.2225	0.157	-1.421	0.155	-0.529	0.084
sugar_stan	0.0958	0.067	1.427	0.154	-0.036	0.227
fat_stan	0.3530	0.106	3.326	0.001	0.145	0.561
chol_stan	-0.0628	0.047	-1.333	0.183	-0.155	0.030
water_stan	0.0682	0.037	1.866	0.062	-0.003	0.140
salt_stan	0.0399	0.066	0.606	0.544	-0.089	0.169
caff_stan	0.0192	0.039	0.488	0.626	-0.058	0.096
prediabetes	0.1017	0.102	0.997	0.319	-0.098	0.302
diabetes	0.0826	0.122	0.679	0.497	-0.156	0.321
private_ins	-0.0441	0.085	-0.520	0.603	-0.210	0.122
rare_drink	-0.0288	0.157	-0.184	0.854	-0.336	0.278
some_drink	-0.1104	0.163	-0.678	0.498	-0.430	0.209
often_drink	0.1395	0.174	0.799	0.424	-0.203	0.481
drk_but_no12	0.2322	0.170	1.366	0.172	-0.101	0.565
fiber	-0.0097	0.006	-1.537	0.124	-0.022	0.003
vit_e	-0.0225	0.009	-2.433	0.015	-0.041	-0.004
mexican	0.0456	0.188	0.242	0.809	-0.323	0.415
hispanic	0.1132	0.160	0.708	0.479	-0.200	0.427
black	-0.5373	0.135	-3.994	0.000	-0.801	-0.274
asian	0.4502	0.206	2.185	0.029	0.046	0.854
other	0.2065	0.158	1.309	0.190	-0.103	0.516
0/1	5.9175	0.409	14.465	0.000	5.116	6.719
1/2	-0.3248	0.048	-6.779	0.000	-0.419	-0.231
2/3	-0.0702	0.043	-1.650	0.099	-0.154	0.013

We will run our model one more time before checking for proportionality, this time without the private_ins covariate, and the kcal covariates. Caffeine has no clinical backing as being a covariate for FLD, showed little to no correlation in the EDA, and has an insignificant p-value here. KCal had an incredibly high VIF in the EDA stage (37!), and is insignificant here. As discussed earlier, KCal is likely a mask for high quantities in other macronutrients and BMI. Hence we have reason to drop them in this upcoming model.

	coef	std err	z	P> z	[0.025	0.975]
cig_smoker_ever	0.1167	0.079	1.476	0.140	-0.038	0.272
sleep_weekday	-0.0087	0.025	-0.345	0.730	-0.058	0.041
no_ins_12_mon	-0.2885	0.189	-1.525	0.127	-0.659	0.082
bmi	0.1940	0.007	25.887	0.000	0.179	0.209
sex	0.3423	0.080	4.278	0.000	0.185	0.499
age	0.0167	0.003	6.231	0.000	0.011	0.022
born_us	-0.1039	0.128	-0.809	0.419	-0.356	0.148
fam_inc_to_pov	-0.0082	0.028	-0.293	0.770	-0.063	0.047
hep_b	0.0800	0.294	0.272	0.785	-0.496	0.656
mins_seden_daily_stan	0.0076	0.039	0.193	0.847	-0.069	0.085
sugar_stan	-0.0522	0.044	-1.195	0.232	-0.138	0.033
fat_stan	0.1250	0.067	1.854	0.064	-0.007	0.257
chol_stan	-0.0162	0.047	-0.346	0.729	-0.108	0.075
water_stan	-0.0293	0.039	-0.745	0.456	-0.106	0.048
salt_stan	0.1790	0.058	3.073	0.002	0.065	0.293
prediabetes	0.2147	0.101	2.122	0.034	0.016	0.413
diabetes	0.0158	0.121	0.130	0.896	-0.222	0.253
private_ins	-0.0457	0.084	-0.542	0.588	-0.211	0.120
rare_drink	-0.0182	0.155	-0.117	0.907	-0.323	0.286
some_drink	-0.1741	0.161	-1.078	0.281	-0.491	0.142
often_drink	-0.1984	0.169	-1.170	0.242	-0.530	0.134
drk_but_no12	0.0055	0.169	0.033	0.974	-0.325	0.336
fiber	-0.0083	0.006	-1.378	0.168	-0.020	0.003
vit_e	-0.0335	0.009	-3.603	0.000	-0.052	-0.015
mexican	0.1637	0.188	0.871	0.384	-0.205	0.532
hispanic	-0.1250	0.160	-0.782	0.434	-0.438	0.188
black	0.0575	0.131	0.440	0.660	-0.199	0.314
asian	0.0768	0.208	0.369	0.712	-0.331	0.485
other	-0.0900	0.157	-0.572	0.567	-0.398	0.218
0/1	5.3935	0.403	13.396	0.000	4.604	6.183
1/2	-0.3321	0.048	-6.931	0.000	-0.426	-0.238
2/3	-0.0810	0.043	-1.901	0.057	-0.164	0.003

This model looks far cleaner than the one we started with, but still has a ways to go. We will

now perform a Brant-Wald Test for proportionality on this model.

Brant-Wald Test Result

	X2	df	probability
0	76.449176	58	0.052672

From this Brant test, parallel regression holds for this model barely, and we will now proceed to testing additional models based off of this model. We will make sure to check this assumption again in the future when we have come to a final model.

Testing Several models

The first model we will test will simply drop all the non-significant covariates from our above model at the $\alpha = 0.05$ level. Note that we keep categorical covariates that are insignificant if they have a category involved that is significant, so as not to treat certain categories within the covariate as 'special'.

This means we will drop the cigarette smoking binary, sleep_weekday, no_ins_12_mon, born_us, fam_inc_to_pov, hep_b, mins_seden_daily_stan, sugar_stan, fat_stan, chol_stan, water_stan, private_ins, all drinking covariates, fiber, and all race covariates.

Backwards Selection Model

	coef	std err	z	P> z	[0.025	0.975]
bmi	0.1931	0.007	26.544	0.000	0.179	0.207
sex	0.2282	0.078	2.943	0.003	0.076	0.380
age	0.0164	0.002	6.984	0.000	0.012	0.021
salt_stan	0.2105	0.043	4.842	0.000	0.125	0.296
prediabetes	0.5368	0.100	5.356	0.000	0.340	0.733
diabetes	0.1993	0.117	1.706	0.088	-0.030	0.428
vit_e	-0.0333	0.007	-4.501	0.000	-0.048	-0.019
0/1	5.7762	0.268	21.541	0.000	5.251	6.302
1/2	-0.3363	0.048	-7.027	0.000	-0.430	-0.242
2/3	-0.0813	0.043	-1.911	0.056	-0.165	0.002

This model performs phenomenally. As no surprise, it contains covariates which were all signaled to be important in the EDA stage, all with low VIF as well.

Our next model will be carefully selected from the previous information and clinical reasoning. It will contain all the covariates from the previous model, as they are statistically and clinically

significant. It will also contain drinking covariates and mins_seden_daily_stan. Basically, we are adding covariates which we know to be clinically significant in liver fat to the model

Clinical Reasoning Model

	coef	std err	z	P> z	[0.025	0.975]
bmi	0.1967	0.007	26.761	0.000	0.182	0.211
age	0.0181	0.002	7.460	0.000	0.013	0.023
sex	0.3604	0.078	4.594	0.000	0.207	0.514
prediabetes	0.1015	0.100	1.013	0.311	-0.095	0.298
diabetes	0.1004	0.119	0.845	0.398	-0.132	0.333
vit_e	-0.0353	0.007	-4.768	0.000	-0.050	-0.021
mins_seden_daily_stan	-0.0202	0.038	-0.530	0.596	-0.095	0.055
salt_stan	0.2111	0.044	4.830	0.000	0.125	0.297
drk_but_no12	-0.3545	0.164	-2.158	0.031	-0.676	-0.033
rare_drink	-0.3100	0.151	-2.053	0.040	-0.606	-0.014
some_drink	-0.1204	0.157	-0.768	0.443	-0.428	0.187
often_drink	-0.3036	0.162	-1.878	0.060	-0.620	0.013
0/1	5.6862	0.300	18.924	0.000	5.097	6.275
1/2	-0.3392	0.048	-7.086	0.000	-0.433	-0.245
2/3	-0.0871	0.043	-2.046	0.041	-0.171	-0.004

We see this model also performs well. diabetes is insignificant, but prediabetes, isn't, a contrast from the last model, one of the covariates may have multicollinearity with the diabetes binary. Overall, this model looks strong with three diet covariates being significant.

Our last model will be created from the EDA only. This means it will include bmi, sex, age, no_ins_12_mon, diabetes, sugar_stan, fiber_stan, fat_stan, and vit_e, all the variables picked out as most important.

EDA Supported Model

	coef	std err	z	P> z	[0.025	0.975]
bmi	0.1954	0.007	26.524	0.000	0.181	0.210
age	0.0151	0.002	6.366	0.000	0.010	0.020
sex	0.1542	0.077	2.005	0.045	0.003	0.305
no_ins_12_mon	-0.4510	0.189	-2.387	0.017	-0.821	-0.081
prediabetes	0.1328	0.100	1.330	0.184	-0.063	0.329
diabetes	0.3007	0.118	2.548	0.011	0.069	0.532
sugar_stan	0.1488	0.042	3.583	0.000	0.067	0.230
fiber	-0.0057	0.005	-1.045	0.296	-0.016	0.005
fat_stan	0.1212	0.050	2.441	0.015	0.024	0.219
vit_e	-0.0275	0.009	-3.013	0.003	-0.045	-0.010
0/1	5.6218	0.280	20.059	0.000	5.073	6.171
1/2	-0.3409	0.048	-7.121	0.000	-0.435	-0.247
2/3	-0.0869	0.043	-2.041	0.041	-0.170	-0.003

We see this model also looks quite strong. Fiber is again insignificant, but diabetes regains its significance interestingly, and no_ins_12_mon is significant as well.

Overall we avoid forward selection of testing and keeping significant p-values as advised in class, and come out with four models from a combination of backwards selection and different types of statistical reasoning. Now we test these four models to

BIC and LR Tables

Below is a table of LR test results performed of the three models against that initial preprop model we used to select them from. It also contains the BIC and AIC value for each model.

LR Test, AIC, and BIC Table

	Models	LR_Test	AIC	BIC
0	Prop Tested Model	NaN	6606.414638	6798.661039
1	Backwards Selected Model	0.983338	6572.718134	6632.795134
2	Clinically Selected Model	0.143697	6595.587810	6685.703310
3	EDA Selected Model	0.141332	6594.027850	6672.127950

As we can see from the table above, our backwards selection model does the best in all testing categories. This makes sense, as the model is selected for the purpose of clinical significance. Our ‘Clinically Selected Model’ and ‘EDA Selected Model’ perform quite similarly, hence I will prefer the ‘Clinically Selected Model’, as it aligns with both the data and clinical reasoning, not just the data. For this reason, the two models we will proceed with into our final analysis will be the ‘Backwards Selected Model’ and the ‘Clinically Selected Model’. While it looks like

we have a clear winner in the ‘Backwards Selected Model’, we will compare both with CV in our final analysis.

Coefficient Interpretations for ‘Best Model’.

Of the two models we narrowed our search down to above, the leading model in my opinion is the ‘Backwards Selection Model’ and its summary table is shown below. Also given is a table of the standardization values for salt needed in the interpretation.

Non-Statistical Explanation of Cumulative Odds Ratio

I will be interpreting the change in cumulative odds. Basically how ones odds of being in the k-th FLD category or below compared to higher than the kth category change when holding all covariates constant and increasing the covariate of interest by 1. The following fraction is the ratio of two cumulative odds for reference...

$$\frac{\frac{P(Y_i \leq k | X_{i,j} + 1)}{P(Y_i > k | X_{i,j} + 1)}}{\frac{P(Y_i \leq k | X_{i,j})}{P(Y_i > k | X_{i,j})}} = e^{-\beta_j}$$

$$k \in (0, 1, 2), i \in (1, \dots, n), j \in (1, \dots, p)$$

We see if $e^{-\beta_j} > 1$, the numerator is greater than the denominator, and hence the cumulative odds of the $X_{i,j} + 1$ are higher than the $X_{i,j}$ case, meaning the increase in X made $P(Y_i \leq k)$ more likely. Hence, $e^{-\beta_j} < 1$ has the opposite effect, it makes $P(Y_i > k)$ more likely. In short, $e^{-\beta_j} < 1$ shrinks the cumulative odds, meaning $P(Y_i \leq k)$ increases, and the covariate associated with β_j has a positive effect on FLD classification.

Backwards Model Table

	coef	std err	z	P> z	[0.025	0.975]
bmi	0.1931	0.007	26.544	0.000	0.179	0.207
sex	0.2282	0.078	2.943	0.003	0.076	0.380
age	0.0164	0.002	6.984	0.000	0.012	0.021
salt_stan	0.2105	0.043	4.842	0.000	0.125	0.296
prediabetes	0.5368	0.100	5.356	0.000	0.340	0.733
diabetes	0.1993	0.117	1.706	0.088	-0.030	0.428
vit_e	-0.0333	0.007	-4.501	0.000	-0.048	-0.019
0/1	5.7762	0.268	21.541	0.000	5.251	6.302
1/2	-0.3363	0.048	-7.027	0.000	-0.430	-0.242
2/3	-0.0813	0.043	-1.911	0.056	-0.165	0.002

Salt Standardization Table (mg)

	Mean	Std. Dev
0	3115.828063	1315.348586

From the above table, we see by taking e^{-coef} , we interpret our coefficients as follows.

Holding all other covariates constant, a one unit increase in bmi scales the cumulative odds by 0.824, meaning an increase in BMI has a positive effect on FLD classification.

Holding all other covariates constant, being male increases the cumulative odds by 0.796, meaning being male over female has a positive effect on FLD classification.

Holding all other covariates constant, a one unit increase in sex scales the cumulative odds by 0.984, meaning an increase in age has a positive effect on FLD classification.

Holding all other covariates constant, a one standard deviation increase in salt intake (mg) scales the cumulative odds by 0.81, meaning an increase in salt intake has a positive effect on FLD classification.

Holding all other covariates constant, being prediabetic over no diabetic condition scales the cumulative odds by 0.585, meaning being prediabetic has a positive effect on FLD classification.

Holding all other covariates constant, being diabetic over no diabetic condition scales the cumulative odds by 0.819, meaning being diabetic has a positive effect on FLD classification.

Holding all other covariates constant, a one unit increase in Vitamin E scales the cumulative odds by 1.034, meaning an increase in Vitamin E intake has a negative effect on FLD classification

As far as interpreting intercepts goes, python does not provide the intercepts in proper form, the a link to an explanation can be found [here](#). I will interpret them now.

For a female with a BMI of 0, age of 0, average salt intake, no diabetic condition, and no vitamin e intake, her being in group 0 is 322.521 times as likely as being an FLD group higher than 0. This makes sense, as no individual would actually have no bmi or age, so intercept interpretation here is not too relevant without individual. We still proceed as follows.

For a female with a BMI of 0, age of 0, average salt intake, no diabetic condition, and no vitamin e intake, her being in group 1 or lower is 658.912 times as likely as being an FLD group higher than 1.

For a female with a BMI of 0, age of 0, average salt intake, no diabetic condition, and no vitamin e intake, her being in group 2 or lower is 1656.543 times as likely as being in FLD group 3.

From this model interpretation we draw the following conclusions, most of which line up with clinical expectations. As age and BMI increase, FLD the probability of someone being in a higher FLD category increases. This makes sense, as age generally worsens health on the whole, and BMI is a mask for body fat, which is going to obviously be highly associated with liver fat. Being male also has a positive effect on FLD categorization, which could be partly genetic, but also lifestyle indicative; men are more likely to drink alcohol and be overweight after all.

Interestingly, prediabetes seems to have a larger effect than diabetes. This could be an issue with a dataset, but I likely suspect it is because of type 1 and type 2 diabetes differences. Someone who is prediabetic is prediabetic type 2, meaning their diabetes is more than likely lifestyle induced. Someone who is type 1 has diabetes which is not lifestyle induced, and therefore has likely managed their potential liver damage, compared to someone who is type 2 who damaged their liver on the way to their diagnosis. It would be interesting to have a diabetes type classifier, but this information was not available.

Diet covariates are discussed more below.

Overall Research Question Connection

As far as my research question goes, we have some interesting results. Alcohol shows some significance in our clinically selected model, but not in our backwards selected one. It will be interesting to see how this performance goes in our HUM comparison.

As far as exploring age goes, we see that obviously an increase in age increases FLD chance. However, its nowhere near as strong as other covariates. This should be a warning sign to young people, you are not invincible, and while your age may protect you slightly, it won't as much as your other habits which cause diseases like diabetes or obesity.

Lastly, in the case of predicting FLD on the whole, we see that bmi, age, and sex are critical, as conventionally thought. Vitamin E also seems to lower FLD risk, which matches up with clinical reasoning, showing the importance of micronutrients as well. Furthermore salt seems to have a positive effect on FLD categorization, but this clinically doesn't make sense. It leads me to believe high salt content is a mask for other macronutrients and diets, ie a high salt diet may be indicative of processed foods and diets with high fat and sugar.

Initial Issues

I also originally had model convergence issues, which was fixed by just increasing the amount of model iterations to 100000, at which point the first model converged well below that number (20000ish), and subsequent ones lower and lower.

Furthermore, Python has no native method to run a Brant-Wald for proportionality, hence I had to port my data over to R, run the same model, and perform the test before moving on.

Statistically, there seem to be some in depth masking issues between diet covariates and exercise. The flip-flopping of significant covariates indicates that there are complex relationships between diet covariates. For example, salt seems significant in our small model, but has no strong clinical significance in fat production. However, a high salt diet is often a sign of a diet high in unhealthy processed foods, so salt may just be the best mask for overall diet. Because of limited time in addressing all the complex relationships between all diet covariates, the backwards selection model provides the best chance of finding truly significant predictors. Basically, without time to unmask every diet predictor, the best outcome can be found by picking the diet covariates with the strongest signal, ie salt. It has a low VIF compared to the clinical diet covariates, like sugar and fat, and likely signals the same message about an unhealthy diet. Another example would be fiber. High fiber has no clinical significance in liver fat, but in the EDA it showed promise, and in some models in this document it did too, but its more than likely just a mask for a healthy lifestyle. In short, the statistical solution for highly correlated diet covariates is to pick the one with the strongest signal for what they all try to convey, which seems to be salt.

Steps for Final Report

I have narrowed myself down to two models using AIC/BIC values and LR tests. I will perform cross validation in the next portion of the project, with HUM as the metric, and use this to decide on my final model. On this final model, I will run another Brant-Wald test, a Pearson Goodness of Fit test, and use cook's distance to find high leverage high residual points and assess whether there is justification to remove them from the model to tighten up estimates without compromising our research process, or whether they should remain. I will finally interpret this final model.

Appendix

Overview

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from statsmodels.miscmodels.ordinal_model import OrderedModel
import scipy.stats as stats

nhanes_df_start = pd.read_csv('./data/final_nhanes_df.csv')
nhanes_df = nhanes_df_start
```

Initial Proportional Model

```
# setting up the initial columns

# renaming other to other_race really quick

nhanes_df = nhanes_df.rename({
    'race': 'other_race'
})

initial_cols = ['cap_cat', 'cig_smoker_ever', 'sleep_weekday', 'no_ins_12_mon', 'bmi', 'sex']

initial_XY = nhanes_df[initial_cols].dropna()


# our initial model with everything

first_model = OrderedModel(
    initial_XY['cap_cat'],
    initial_XY.drop(columns = ['cap_cat']),
    distr = 'logit'
).fit(maxiter = 100000, disp = False)

display(first_model.summary().tables[1])


# our next model without carbs, kcal, and education level

next_XY = initial_XY.drop(columns = ['carb_stan', 'hs_dropout', 'hs_grad', 'aa', 'bach'])

next_model = OrderedModel(
    next_XY['cap_cat'],
    next_XY.drop(columns = 'cap_cat'),
    distr = 'logit'
).fit(maxiter = 100000, disp = False)

display(next_model.summary().tables[1])


# model before proportionality test

preprop_XY = next_XY.drop(columns = ['kcal_stan', 'caff_stan'])

preprop_XY_model = preprop_XY
```

```

preprop_model = OrderedModel(
    preprop_XY_model['cap_cat'],
    preprop_XY_model.drop(columns = 'cap_cat'),
    distr = 'logit'
).fit(maxiter = 100000, disp = False)

display(preprop_model.summary().tables[1])

```

```

# exporting df for prop test

preprop_XY.to_csv('./data/preprop_XY.csv', index = False)

```

```

# showing the results of Brant-Walt Test from R

initial_prop_test = pd.read_csv('./data/first_prop_result.csv')
display(initial_prop_test)

```

Testing Several Models

```

# backward selection model

XY1 = preprop_XY.drop(columns = ['cig_smoker_ever', 'sleep_weekday', 'no_ins_12_mon', 'born_

model_1_no_fit = OrderedModel(
    XY1['cap_cat'],
    XY1.drop(columns = 'cap_cat'),
    distr = 'logit'
)

model_1 = model_1_no_fit.fit(maxiter = 100000, disp = False)

display(model_1.summary().tables[1])

```

```

# clinical reasoning model

XY2 = preprop_XY[['bmi', 'age', 'sex', 'prediabetes', 'diabetes', 'vit_e', 'cap_cat', 'mins_

model_2 = OrderedModel(
    XY2['cap_cat'],

```

```

    XY2.drop(columns = 'cap_cat'),
    distr = 'logit'
).fit(maxiter = 100000, disp = False)

display(model_2.summary().tables[1])

```

```

# EDA model

```

```

XY3 = preprop_XY[['bmi', 'age', 'sex', 'no_ins_12_mon', 'prediabetes', 'diabetes', 'sugar_st

model_3 = OrderedModel(
    XY3['cap_cat'],
    XY3.drop(columns = 'cap_cat'),
    distr = 'logit'
).fit(maxiter = 100000, disp = False)

display(model_3.summary().tables[1])

```

```

# making table with aic, bic, and LR values

```

```

model_names = ['Prop Tested Model', 'Backwards Selected Model', 'Clinically Selected Model',

bic_models = [preprop_model.bic, model_1.bic, model_2.bic, model_3.bic]
aic_models = [preprop_model.aic, model_1.aic, model_2.aic, model_3.aic]

def lr_ordered_model(full_model, nested_model):
    lr = 2 * (full_model.llf - nested_model.llf)
    df = full_model.params.size - nested_model.params.size
    p = 1 - stats.chi2.cdf(lr, df)
    return(p)

lr_test_ps = [np.nan, lr_ordered_model(preprop_model, model_1), lr_ordered_model(preprop_model, model_2), lr_ordered_model(preprop_model, model_3)]

compare_df = pd.DataFrame({
    'Models': model_names,
    'LR_Test': lr_test_ps,
    'AIC': aic_models,
    'BIC': bic_models
})

display(compare_df)

```

```

# creating salt standardization table and displaying it with backwards selection model summary

salt_mean = nhanes_df['salt'].mean()
salt_std = nhanes_df['salt'].std()

salt_table = pd.DataFrame({
    'Mean': [salt_mean],
    'Std. Dev': [salt_std]
})

display(model_1.summary().tables[1])
print('Salt Standardization Table (mg)')
display(salt_table)

intercepts = model_1_no_fit.transform_threshold_params(model_1.params)

```

R Code in Seperate File

```

# | eval: false

# performing prop test in R

library(tidyverse)
library(MASS)
library(brant)

r_prepropXY = read_csv('./data/preprop_XY.csv')
r_prepropXY$cap_cat = ordered(r_prepropXY$cap_cat)

r_preprop_model <- polr(
  cap_cat ~ .- cap_cat,
  data = r_prepropXY,
  method = 'logistic',
  control = list(maxit = 100000)
)

result <- brant(r_preprop_model)
brant_df <- as.data.frame(result)
total_prop_result = brant_df[1,]
write.csv(total_prop_result, './data/first_prop_result.csv', row.names = FALSE)

```